Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>
the de Bro gl ie wavelength is given by the equation Î» = h / âĪļ (2 m E), where h is the Plan ck constant , m is the mass of the electron , and E is the kinetic energy .
first , convert the kinetic energy from e V to Jou les : 1 e V = 1 . 6 ÃĹ 10 ^- 19 J .
then , the kinetic energy in Jou les is 1 ÃĹ 1 . 6 ÃĹ 10 ^- 19 J = 1 . 6 ÃĹ 10 ^- 19 J .
the mass of an electron is approximately 9 . 11 ÃĹ 10 ^- 31 kg .
the Plan ck constant is approximately 6 . 626 ÃĹ 10 ^- 34 J s .
now , plug in the values into the equation : Î» = (6 . 626 ÃĹ 10 ^- 34 J s) / âĪļ (2 ÃĹ 9 . 11 ÃĹ 10 ^- 31 kg ÃĹ 1 . 6 ÃĹ 10 ^- 19 J).
first , calculate the denominator : âĪļ (2 ÃĹ 9 . 11 ÃĹ 10 ^- 31 kg ÃĹ 1 . 6 ÃĹ 10 ^- 19 J) = âĪļ (2 . 912 ÃĹ 10 ^- 49 kg J).
then , calculate the square root : âĪļ (2 . 912 ÃĹ 10 ^- 49 kg J) = 1 . 705 ÃĹ 10 ^- 24 kg m / s .
now , divide the numerator by the square root of the denominator : Î» = (6 . 626 ÃĹ 10 ^- 34 J s) / (1 . 705 ÃĹ 10 ^- 24 kg m / s).
finally , calculate the result : Î» = 3 . 88 ÃĹ 10 ^- 10 m .
convert the result to cm : Î» = 3 . 88 ÃĹ 10 ^- 10 m ÃĹ (100 cm / 1 m) = 3 . 88 ÃĹ 10 ^- 8 cm .
</think><answer> A . 2 . 48 ÃĹ 10 ^- 8 cm